My Amazing Thesis Title

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Submitted to the

Distinguished Majors Program

Department of Economics
University of Virginia

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This is the abstract of my thesis. Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip ex ea commodo consequat.

Keywords: machine learning, algorithms, optimization

JEL-Classification: C61, C63

Email: name@lastname.com (use an email that will be accessible after graduation).. I would like to thank my advisor and family...

1. Introduction

Use of latex, rather than Word, is highly recommended for your thesis. Here is a short summary of their comparison and the advantages of latex:

https://blog.orvium.io/latex-over-word/

You have the option to keep using Word, but you have to follow the same format (next section) as this template. Also, all thesis must have the first two title pages in the same way regardless of the writing software used.

- This template is adapted from Kyle Butt's template available in https://github.com/kylebutts/latex-templates which also provides a latex beamer template for slides.
- The other file in this folder, `thesistemplate.tex`, is a simple version with only a skeleton structure.

  `titlepage.tex` is used in the template to insert the title page and academic page, and is not designed to be compiled on its own. All the information in pages 1 and 2 are set in the preamble of main .tex file. `paper.sty` is a style file that has a set of packages and commands that are useful for writing papers. Note that both style files (`paper.sty` and `math.sty`) need to be in the same folder as this main.tex file.[1]
- the bibliography file `references.bib` is in BibTex format. You can export your references from Google Scholar, Zotero, Mendeley, or other reference management software in BibTex format and add them to this file. It's loaded at the end by the `bibliography` command.
- To keep things organized, I recommend you create folders named `figures` and `tables` to store your figures and tables, respectively. You can then call them from that path when needed. See below for examples.

To get this all going, you need a latex compiler and a latex editor. For Mac, I recommend TexShop as an editor and MacTex as compiler free at `https://www.tug.org/mactex/`. Visual Studio Code also works, or you can do it online via Overleaf where you have UVA license: go to `https://www.overleaf.com/`, start a project, and upload all the contents of this folder. You don't need to download and install any compiler and editor, and can access it from any computer!

Overleaf also has a very good tutorial for beginners:

1. The file `math.sty` has a set of useful math operators. Credit to `https://pascalmichaillat.org/d3/`.

https://www.overleaf.com/learn/latex/Learn_LaTeX_in_30_minutes

1.1. Format and Margins

The paper is set to 12pt Sans-serif font, with 1.5 line spacing, and 1 inch margins. Do not temper with these defaults. You can personalize some aspects of the style file, but do not change the margins, font character, font size, or line spacing.

1.2. Some Commands

See below for Theorem 1, the regression specification (1), Table 1, Figure 1, appendix Table Appendix A1. Of course, make sure to touch up on your micro theory with Mas-Colell et al. (1995).

Let's cite something in parenthesis: (Barjamovic et al., 2019), or multiple people in parenthesis, as in: (Mas-Colell et al., 1995; Barjamovic et al., 2019)

You can make a word bold with **bold text** or italicized with *italicized text*.

1.3. Compiling

If you use your own compiler, getting the references and links to sections and equations to work requires multiple compilations. If you use Overleaf, it does this automatically for you.

In some compilers, you may need to run the following commands in sequence: first, run `pdflatex` to compile the main document, then run `bibtex` to compile the bibliography, and finally run `pdflatex` twice to ensure that all references and links are correctly updated.

2. Highlights

2.1. Math Commands

Theorem environment: there are the following environments: `theorem`, `proposition`, `assumption`, `example`, `lemma`, `corollary`, `definition`, `remark` – may not be super relevant for empirical work though.

Theorem 1 (Example Theorem)**.** This is an example theorem

$$\hat{\beta} = \frac{\sum_\ell e_\ell z_\ell y_\ell^\perp}{\sum_\ell e_\ell z_\ell x_\ell^\perp}$$

Jibberish math to show off symbols:

$$y = f(X) + \varepsilon = X\beta + \psi_i + \nu_t + w_{i,t} + \varepsilon_{i,t} \tag{1}$$

When you use the `equation` command, you get numbered equations that you can label and refer to such as equation 1 – as you change things, this updates automatically.

The command $$ $$ with math in between creates unnumbered equations, as does this:

$$y = x$$

The command $ $ creates in line math: $y = z$

`\one` does an indicator (dummy variables!):

$$1\left[X_i > 0\right]$$

2.2. Tables

For tables, I use the `tabular` and `booktabs` packages. For table and figure notes, I use a custom `\note` command. It uses `\parbox` under the hood. You can use it in one of of four ways:

1. `\note{text}`
2. `\note[Notes.]{text}`
3. `\note{0.6\textwidth}{text}`
4. `\note[Notes.]{0.6\textwidth}{text}`

In addition, I use the `adjustbox` package for resizing figures/tables. It automatically scales the figure/table proportionally, so things look right. For example, here's a table that's too wide. I use the `adjustbox` package to fix it.

Figures also use the `\note` and the `adjustbox` package. Here's an example figure:

In practice, `tikzpicture` environment is not super important for your purposes but it has conveniences depending on the need. You will mostly use:

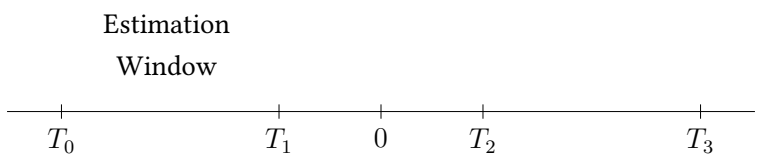
```
\begin{figure}[htbp]
    \centering
```

Table 1 — Table Too Wide (adjustbox)

			Market Access		Urban Weekly Wage		Nonurban Weekly Wage	
Year	N	% Urban	Mean	SD	Mean	SD	Mean	SD
1940	16,875,829	0.66	10,708.21	14,819.55	33.22	19.66	25.23	16.56
1950	67,790	0.69	23,166.06	26,600.85	70.05	32.70	58.27	29.24
1960	1,338,491	0.66	40,328.17	45,385.47	124.11	77.55	99.37	59.01
1970	277,951	0.75	65,373.07	68,048.88	205.42	137.80	165.61	104.49
1980	1,907,836	0.73	73,223.43	74,906.62	408.83	284.44	340.82	220.68
1990	2,257,874	0.68	224,312.50	168,933.04	711.77	574.58	550.26	392.90
2000	2,581,741	0.73	288,195.77	199,100.92	1,010.36	1,022.77	754.10	649.57
2010	530,359	0.76	273,754.31	195,220.04	1,306.70	1,281.34	972.40	817.94

Notes. Weekly wage is reported in 2015.

Figure 1 — Event-timing



Notes. This is an example figure in the paper

```
    %\includegraphics[width=0.8\textwidth]{graph.pdf}
    \caption{This is the figure caption describing what the figure shows.}
    \label{fig:example}
\end{figure}
```

[remove the verbatim commands to make this a real command that renders `graph.pdf` file] You display the file titled `graph.pdf` with the appropriate path structure, i.e., it can be in a folder titled `figures` and used as

```
\includegraphics[width=0.8\textwidth]{figures/graph.pdf}
```

For example:

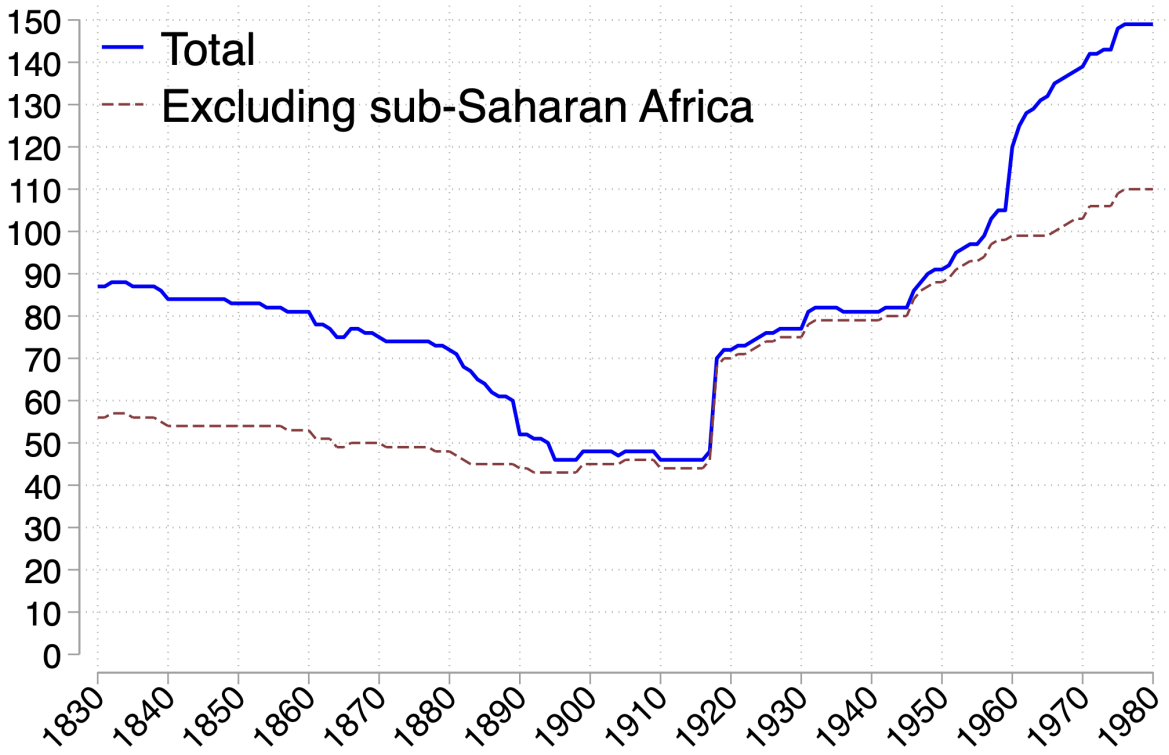


Figure 2 — This is the figure caption describing what the figure shows.

References

Barjamovic, Gojko, Thomas Chaney, Kerem Coşar, and Ali Hortaçsu, "Trade, Merchants, and the Lost Cities of the Bronze Age*," *The Quarterly Journal of Economics*, 03 2019, *134* (3), 1455–1503.

Mas-Colell, Andreu, Michael Dennis Whinston, and Jerry R Green, *Microeconomic theory*, Vol. 1, Oxford university press New York, 1995.

Appendix A. Additional Results

The appendix will automatically start numbering tables, figures, and theorem-like environments using the appendix section \Alph (e.g. Table Appendix A1).

You can still cite Mas-Colell et al. (1995)

Appendix A.1. Regression Results

The main regression equation is

$$y_i = \boldsymbol{X}_i \beta + u_i. \tag{Appendix A.1}$$

Results are presented in Table Appendix A1. In this table example, the table is narrower than textwidth, so I adjust the `\note` width.

Table Appendix A1 — Regression Results

	Dependent variable: Overall Rating	
	(1)	(2)
Handling of Complaints	0.692*** (0.149)	0.682*** (0.129)
No Special Privileges	−0.104 (0.135)	−0.103 (0.129)
Opportunity to Learn	0.249 (0.160)	0.238* (0.139)
Performance-Based Raises	−0.033 (0.202)	
Too Critical	0.015 (0.147)	
Advancement	11.011 (11.704)	11.258 (7.318)
Observations	30	30
Adjusted R^2	0.656	0.682

Notes. Using R base dataframe attitude. dolor in reprehenderit in voluptate velit esse cillum dolore eu fugiat nulla pariatur. Excepteur sint occaecat cupidatat non proident, sunt in culpa qui officia deserunt mollit anim id est laborum.

* $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$.